

M3: Complete Implementation — Unsupervised Detection of Attack Surface Drift

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Abstract—This report details the functional completion of the M3 implementation for detecting Attack Surface Drift. Building on the DBSCAN foundation from M2, I have transitioned to an end-to-end pipeline that incorporates temporal windowing and multi-model anomaly detection. Utilizing an 80,000-event sample systematically streamed from the full LANL authentication telemetry (*auth.txt.gz*), I implemented an Isolation Forest for statistical ranking and a Decision Tree to extract readable rules from behavioral clusters. The results demonstrate the ability to identify “cluster hopping” as accounts migrate from normal baselines into noise categories across the temporal window. This implementation achieves the technical goals for scale and prepares the project for final qualitative assessment in M4.

I. IMPLEMENTATION STATUS

This milestone extends the M2 pipeline into a fully functional end-to-end implementation. The core structure remains intact—including per-user aggregation, standardization, and PCA—but M3 adds the analytical layers required to detect drift rather than just static clusters. I introduced temporal windowing to facilitate early-period versus late-period behavioral comparisons and integrated an Isolation Forest to replace the rule-based drift proxy used in M2.

The implementation has been scaled to process the source telemetry from the LANL dataset. By streaming and sampling from the first 50 million lines of the *auth.txt.gz* file, the pipeline now operates on approximately 80,000 events. This provides the statistical density necessary to validate behavioral clusters across a multi-day temporal window, ensuring that the patterns discovered are representative of the network’s early-state operations and behavioral shifts.

II. ANALYTICAL EXPANSION

The expansion for M3 focused on moving from a single snapshot to a time-dependent discovery process.

A. Temporal Windowing

A major limitation in M2 was the static treatment of logs, which smoothed over behavioral spikes. For M3, I implemented an early/late split based on the median timestamp of the sampled logs. By recomputing features for each sub-period, the model can now measure the behavioral “delta,” such as sudden changes in authentication failures or destination diversity, which serves as the primary indicator of Attack Surface Drift.

B. PCA and Dimensionality Reduction

I reused PCA as a preprocessing step to compress the primary behavioral features into two principal components. This reduced the noise in the high-dimensional telemetry and provided a clear 2D space for the anomaly detection models to operate. The first two components captured approximately 68% of the variance, preserving the structural separation between normal users and irregular accounts.

C. Isolation Forest for Anomaly Detection

I replaced the M2 rule-based proxy with an Isolation Forest. Setting a contamination parameter of 0.1, the model successfully assigned anomaly scores to the profiles. The highest-scoring outliers showed significant overlap with the DBSCAN noise points (*label* = -1), providing cross-model verification that the irregular signatures are statistically robust across the sampled dataset.

D. Decision Tree for Cluster Interpretation

To address the interpretability requirement, I utilized a shallow Decision Tree with the DBSCAN cluster labels as the target classes. This produced a human-readable rule set, showing that destination density (*dst_per_event*) and temporal irregularities (*off_hours_rate*) are the primary features driving the separation of drift accounts from the organizational baseline.

III. RESULTS AND DISCOVERY QUESTION PROGRESS

A. DQ1 — Natural Behavioral Clusters

The four behavioral archetypes (Normal, Service, Admin, and Drift) identified in M2 remain stable in the M3 pipeline. The PCA visualization continues to show a dense central mass for normal activity, while the Isolation Forest scores highlight the irregular accounts pulling away on the failure and protocol axes.

B. DQ2 — Behavioral Drift Over Time

The temporal split identified a subset of target accounts where behavior shifted significantly between the early and late periods. Several users moved from stable clusters into the noise category as their failure rates increased. This “cluster hopping” validates the pipeline’s ability to delimit drift as it happens across the authentication telemetry, rather than just identifying static outliers.

IV. UPDATED M4 PLAN

The technical pipeline is now functionally complete at scale. The final milestone will focus on a qualitative validation of the drift candidates identified in this phase. I will trace these candidates back to the raw LANL events to determine the specific causes of their behavioral shifts. Additionally, I will perform a final sensitivity analysis on the DBSCAN epsilon parameter and explore the inclusion of graph-based features to further refine the separation of unauthorized configurations from administrative noise.

REFERENCES

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